Comparing Bspline and uniGradICON-AI Foundational Deformable Image Registration Methods: Evaluating Tradeoff Between Accuracy and Speed.

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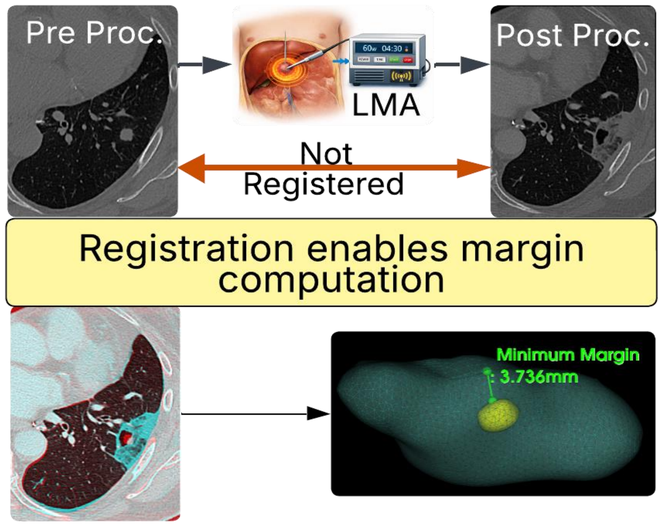
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Motivation

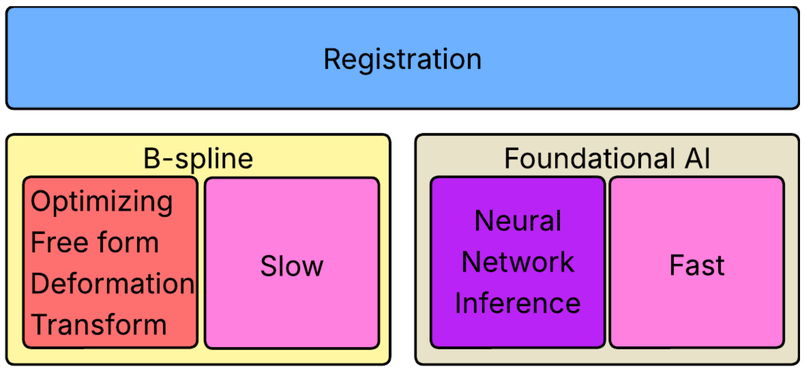
- **Need:** Accurate AND fast deformable CT registration is critical for ablation margin assessment in Lung Microwave Ablation (LMA).
- **Methods:** Classical B-spline vs Foundational AI registration
- **Goal:** Comparison of B-splines and uniGradICON for LMA CT registration for evaluation of accuracy vs speed

INTRODUCTION

LMA Registration

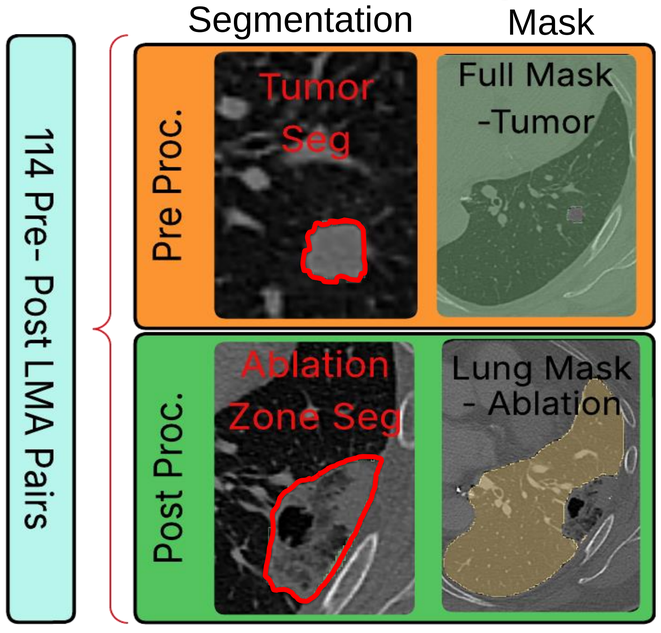


Bspline vs Foundational:

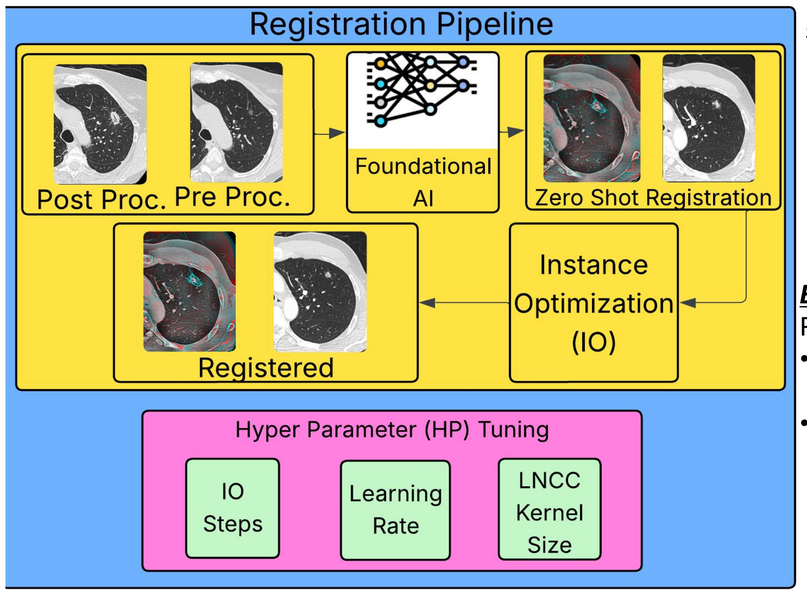


Materials and Methods

Dataset:



Materials and Methods



Setup:
- **uniGradICON** (default; optimized; zero-shot + IO)
- **IO:** Pair specific Loss optimization beyond zero shot inference
- **Loss Function Masking**
- **Dataset-specific Hyper Parameter grid search**

Evaluation Metrics

Registration Accuracy
- **Primary metric**: Target Registration Error (TRE).
- **Landmarks**: Vessel bifurcations annotated by a board-certified radiologist.

Results

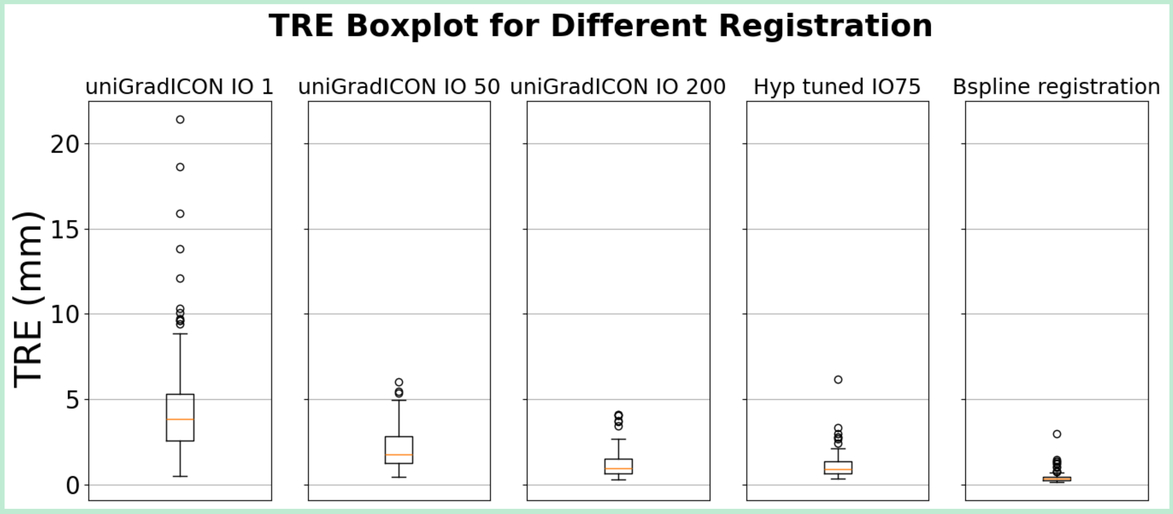


Figure 1. Target registration error (TRE) for uniGradICON Different IO vs B-spline

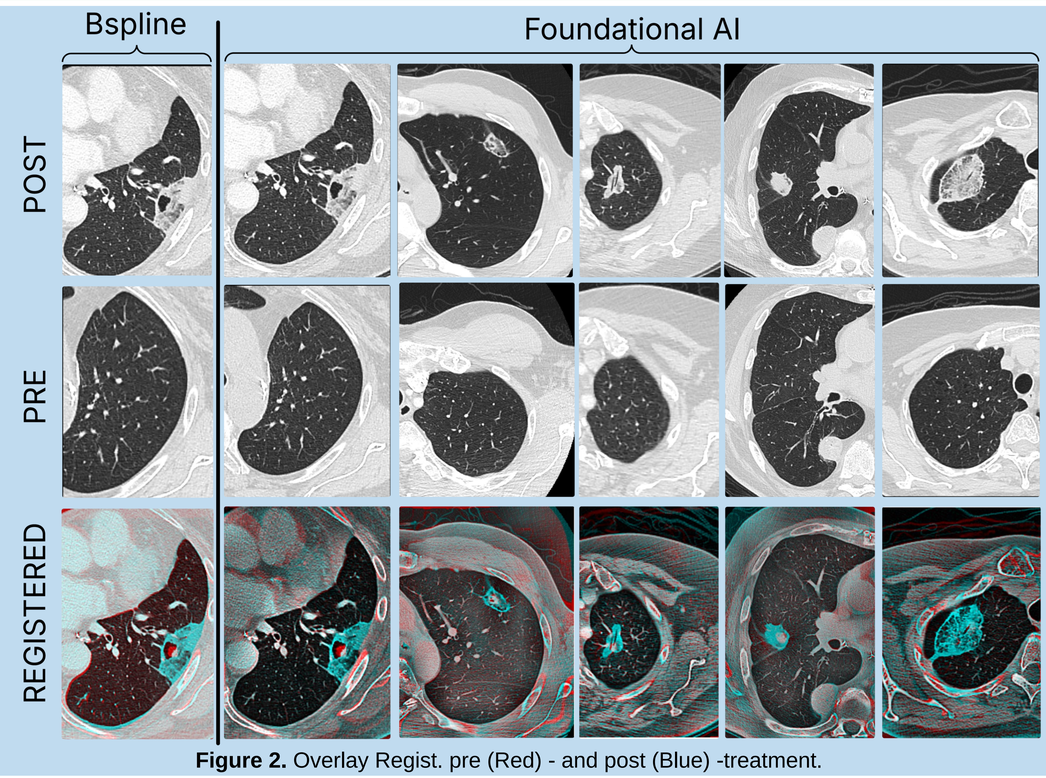


Figure 2. Overlay Regist. pre (Red) - and post (Blue) -treatment.

Conclusion

- **B-spline** achieves **lowest TRE** of 0.43mm (highest accuracy) vs 1.41 mm uniGradIcon foundational AI registration.
- B-spline is **10× slower** than uniGradICON.
- **Instance optimization (IO)** and **hyperparameter tuning** improve accuracy for uniGradICON.
- Results show **accuracy–speed tradeoff** between classical and foundational AI Registration .

Discussion

- Difference in performance may be influenced by uniGradICON's internal downsampling because of memory constraints, requiring further exploration.
- **Domain-specific fine-tuning** may further improve uniGradICON's performance without sacrificing speed.

References

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